

Project 5

TITLE: Smart Home Environment & Security Monitoring System

OBJECTIVE:

To develop a complete IoT mini project combining environmental monitoring, security detection, and alert generation using ESP32.

COMPONENTS USED:

- * ESP32
- * DHT22 Sensor
- * PIR Motion Sensor
- * Red LED
- * Green LED
- * Buzzer
- * Wokwi Simulator

SOFTWARE USED:

- * Arduino IDE
- * Wokwi Simulator

WORKING PRINCIPLE:

The DHT22 sensor monitors temperature while the PIR sensor detects motion. Based on sensor inputs, the ESP32 determines system status and activates LEDs and alarms. The system categorizes situations as Safe, High Temperature, Intruder Detected, or Critical Alert.

FEATURES:

- * Temperature monitoring
- * Motion detection
- * Intruder alert system
- * High temperature warning
- * Critical alert mode
- * LED and buzzer indication

RESULT:

The system successfully integrated multiple sensors and generated appropriate alerts based on environmental and security conditions.

CONCLUSION:

This project combined concepts learned from previous projects and demonstrated a complete IoT workflow involving sensing, decision-making, and alert generation.